Anant Khanna

itsanantk@gmail.com | itsanantk.github.io | linkedin.com/in/itsanantk | github.com/itsanantk

Education

University of British Columbia — Vancouver, BC
Bachelor of Applied Science in ***Computer Engineering*** *- Dean's List 2025* — *Sept. 2024 – May 2028*

Experience

Software Engineer | *Agriprix* — May 2026 – Present

- Rebuilt the hybrid intent router (deterministic rules + **LLM** classification) for an agent platform spanning **70+ tools**; added per-call token/latency **observability** with automatic failover across **5 model providers**
- Implemented **Stripe** authorize-and-capture checkout with deterministic **idempotency** keys, an integer-cent commission pricing engine, and per-farm fulfillment **state machines** for multi-supplier orders
- Led a platform security audit surfacing **67 vulnerabilities (5 critical)**: patched mass-assignment/**IDOR** serializers, eliminated overselling races via **row-level locking**, and shipped a **15-case Playwright** E2E suite covering the supplier-to-buyer order journey
- Built a **WhatsApp** AI assistant on the WhatsApp Business Cloud API, routing inbound webhooks into the agent layer so buyers and suppliers can transact directly over chat

Machine Learning Software Engineer Team Lead | *UBC UAS* — Sept. 2024 – Present

- Integrated **YOLOv11 (PyTorch)** pipeline in Python and C++, enabling real-time **autonomous object detection** on embedded systems travelling at 50+ km/h, **optimizing inference** for low-latency execution
- Engineered a **metric learning pipeline** for target re-identification, computing mean vectors from Vision Transformer **(ViT)** embeddings to enable cosine similarity matching invariant to lighting and orientation
- Implemented **TensorRT FP16 quantization** to compress model weights, reducing memory bandwidth usage by **50%** while maintaining inference precision within **0.5%** of the baseline
- Achieved **98%** detection accuracy with model optimization, improving **mAP by 30%**

Co-founder & Lead Developer | *UBC Finds* — Oct. 2025 – Present

- Built a campus utility finder map on **React/Next.js** and **Supabase**, architecting event-driven state synchronization over Supabase realtime channels that eliminates race conditions through atomic writes
- Optimized report aggregation and live map rendering for low-latency updates as new utility reports stream in

Projects

Wealth Council | *Python, FastAPI, React, Google Gemini, Fetch.ai* — Mar. 2026

- Engineered an agentic investment platform, winning an award at the ProduHacks x Fetch.ai hackathon
- Architected a Multi-Agent Orchestration Layer using **Fetch.ai uAgents** to asynchronously compute portfolio risk metrics and integrated FinBERT **(NLP)** to perform sentiment analysis on live financial news streams
- Integrated Google Gemini to synthesize quantitative modeling and alternative asset correlations, streaming final reports to a React frontend in real-time via **Server-Sent Events (SSE)**

GuardCam: AI Dashcam | *Python, React Native, WebSockets, OpenCV* — Mar. 2026

- Developed an edge-to-cloud driver monitoring system to detect micro-sleeps, utilizing a custom **TCP WebSocket bridge** to bypass HTTP overhead for low-latency video streaming
- Deployed a headless AI server on Modal GPUs to decode base64 frame data in real-time from the edge device
- Implemented **MediaPipe** and **OpenCV** to map a 468-point 3D face-mesh, calculating Eye Aspect Ratio (EAR) to trigger sub-second haptic JSON alerts

FPGA-Based Neural Network Inference Accelerator | *Verilog, FPGA, C++, ARM* — July 2025

- Designed and implemented a **single-layer neural network** (**MAC unit**) on a **DE1-SoC FPGA** using **Verilog**
- Developed a **memory-mapped C++ driver** on the ARM HPS for low-latency data streaming, offloading compute to the FPGA and enabling efficient hardware acceleration under **embedded Linux**
- Achieved **3,000** inferences/sec with a **300%** performance speedup over ARM-only execution by optimizing **HPS–FPGA** communication

Technical Skills

Hardware/RTL: FPGA Design (Intel Quartus, QuestaSim, Platform Designer), Verilog, SystemVerilog, ARM SoCs
Languages: Python, C, C++, C#, SQL, Java, JavaScript, TypeScript, RISC-V & ARM Assembly
Frameworks: Django/DRF, FastAPI, Celery, React, Next.js, PyTorch, TensorFlow, YOLO/Ultralytics
Tools & Cloud: Google Cloud Platform, Docker, PostgreSQL, MySQL, Supabase, Stripe, Playwright, Git, Linux